

Aggregating Information and Preferences under Different Coordination Ability

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Abstract

We investigate majority voting where agents possess private information about an unobservable ground truth that determines their preferences. In such settings, agents may hold different preferences and (collectively) engage in counterintuitive strategic behaviors. Previous work either assumes strategic behavior occurs without coordination or with unlimited coordination, yielding overly inclusive or exclusive predictions about voting outcomes.

We incorporate *coordination ability*—the largest coalition size at which agents could strategically coordinate—into the analysis. Under the ex-ante Bayesian k -strong equilibrium [30] framework, where no group of at most k agents can benefit from deviation, we provide closed-form characterizations of when informed majority decisions, the decision favored by the majority if the ground truth is common knowledge, are achievable. Specifically, we determine (1) when all k -strong equilibria reach the informed majority decision and (2) when at least one such equilibrium exists. These conditions depend on three factors: coordination ability, fraction of majority agents, and information structure. The boundary for the second question exhibits surprising complexity—non-continuous, non-linear, and segmental. Our results reveal the complicated landscape and provide refined predictions for strategic behavior across different coordination levels.

CCS Concepts

• Theory of computation → Algorithmic game theory.

Keywords

Computational Social Choice, Algorithmic Game Theory

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1 Introduction

Majority voting is the most studied and applied method for making efficient, fair, and informed decisions between two alternatives. In practice, it is used in diverse scenarios such as political elections, corporate board meetings, academic or industry hiring processes, and even popular shows, just to name a few. Theoretically, extensive research in social choice supports majority voting as reliable both procedurally and in its conclusions. When agents have complete information about their preferences, majority voting ensures that casting a dishonest vote does not provide an advantage. In scenarios where agents have incomplete information, such as a jury system, the famous Condorcet Jury Theorem [13] shows that agents can vote to collectively reveal the hidden ground truth determining their preferences and reach the “correct” decision with high probability.

However, this observation of the wisdom of crowds rests upon two restrictive assumptions. The Jury Theorem assumes that agents share identical objectives and vote informatively, i.e., honestly reflecting their information in their votes. These assumptions undermine the predictive power of the Jury Theorem in more realistic settings, as illustrated in the following example.

EXAMPLE 1. Consider a legislative vote on a mixed economic policy with uncertain consequences. The policy may produce either inflation or deflation. Representatives come from different social backgrounds and have different preferences. Doves prefer inflation to stimulate economic growth, while Hawks prefer deflation to avoid market bubbles. Each representative possesses private information about the policy’s likely effects from different sources. They may also coordinate with others sharing similar objectives to increase their collective influence. How can majority voting produce efficient, fair, and informed decisions under these conditions?

When agents have divergent preferences, the notion of a universal “correct” decision vanishes. In such cases, a “good” decision should be both fair and efficient, properly aggregating information and preferences from all participants. We adopt the *informed majority decision* [21, 29, 48], the outcome preferred by the majority if agents possessed complete information about their preferences,

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as our benchmark for a desirable outcome. It provides a natural normative criterion that considers both majority preferences and private information. Our central concern is identifying when and how majority voting achieves this informed majority decision.

Strategic considerations introduce surprisingly complex dynamics that cause the Condorcet Jury Theorem to fail. Game-theoretic analysis [6] reveals that informative voting (honestly reflecting private information in the vote) fails to constitute a Nash equilibrium (a prediction of the stable outcome strategic agents will perform), even when all agents share common objectives. A strategic agent, conditioning on being pivotal—the only scenario where their vote matters—can infer information from others' votes that alters their optimal choice. This inference may lead them to vote against their own information. The analysis demonstrates that strategic behavior has a significant effect on the voting behaviors and outcomes, and naturally raises the question. Can informed decisions emerge from majority voting among strategic agents?

A series of previous works aim to address this question and study whether equilibria in the majority vote lead to the informed majority decision. Based on the solution concept they adopted, these works fall into two branches. The first line of works [6, 21, 42, 50] adopts the classical (Bayesian) Nash equilibrium [32], which characterizes stable outcomes where no single agent can benefit by unilaterally changing their strategy. The second, more recent line of works [15, 29, 48] adopts the approximated version of strong Bayes-Nash equilibrium, which requires that no coalition of any size can collectively benefit by jointly changing their votes. However, both solution concepts imposed implicit assumptions about the *coordination ability*—how large a group of agents can coordinate on their behaviors. A Nash equilibrium ignores the potential coordination of strategic agents to break an equilibrium, especially in a voting scenario where a single vote can rarely flip the outcome. A strong equilibrium, on the other hand, is too pessimistic to admit a stable outcome only if it is robust to unlimited-size coordination.

Despite these extreme assumptions in the previous solution concepts, the coordination ability of agents varies depending on the society of the vote. In practice, coordination occurs within bounded groups, such as in local communities or trusted networks, rather than across entire populations. Theoretical and empirical studies indicate that the magnitude of group coordination is restricted by cognitive burden [18], communication complexity [12], and efficiency loss [36], etc. Consequently, the Bayesian Nash equilibrium is too inclusive in its predictions for the stable outcomes of strategic agents, while the strong Bayesian is too exclusive in its predictions. This choice between “no coordination” and “unbounded coordination” is too coarse to characterize the strategic behavior of agents under societies of different abilities to coordinate. Consequently, the following research question remains largely open.

When and how is the informed majority decision reached under societies with different abilities to coordinate?

To address such a question, a more fine-grained solution concept parameterized by the group coordination ability is required. In this paper, we employ the approximated *ex-ante Bayesian k -strong equilibrium* [30] (abbreviated as *k -strong equilibrium*) parameterized by the coordination ability parameter k . In a k -strong equilibrium, no coalition of at most k agents can collectively benefit by jointly

changing their votes. Such a solution concept encompasses both Bayesian Nash equilibrium ($k = 1$) and strong Bayes-Nash equilibrium ($k = n$, n is the number of agents) as special cases and allows conducting a spectrum of equilibrium analyses on different k , fitting into societies with different coordination ability.

1.1 Our Contribution

We address this question by conducting equilibrium analysis on the *ex-ante Bayesian k -strong equilibrium*, which enables us to make more refined predictions on the stable outcomes of strategic behaviors. Our research question can be interpreted into the following two questions.

- (1) When do all k -strong equilibria reach the informed majority decision?
- (2) When does there exist a k -strong equilibrium reaching the informed decision?

The two questions ask whether the worst-case/best-case equilibrium can still reach a good outcome. At a high level, they correspond to the price of anarchy [38] and the price of stability [4], respectively. Under a representative scenario of Example 1 where two groups of agents possess different preferences, we give the closed-form threshold on when each question leads to an affirmative answer, as in Figure 1. Consequently, when we have knowledge on the coordination ability of a voting society, our analysis allows us to make a more refined prediction of the outcome of strategic votes in the worst and the best case.

For the first question, the answer is affirmative if and only if the coordination ability $k \geq \frac{n}{2}$, where n is the number of agents (Theorem 1). This threshold is inevitable, as unanimous votes on the same alternative cannot be flipped by any coalition smaller than a half of the population and becomes an $\frac{n}{2}$ -strong equilibrium. However, unanimous voting ignores all private information and fails to reach informed decisions.

The second question, on the other hand, yields a far more complex answer. Whether such a k -equilibrium exists depends on three factors: the coordination ability k , the fraction α of the agents whose preferences are the majority, and the distribution of agents' private information (Theorem 2). As illustrated in Figure 1, the boundary between existence and non-existence displays non-linear, non-continuous, and non-strictly monotone segmental behavior across different fractions α , revealing the intricate strategic landscape shaped by different levels of coordination ability. In the order as the majority fraction α decreases, it starts with a range of α where a strong equilibrium exists and reaches the informed majority decision. The threshold drops from this maximal value after a critical point and displays at most four distinct segments. Segment 1 exhibits a constant threshold despite decreasing α . Segment 2 is a linear decline in the threshold as α decreases. Segment 3 reveals the most complex behavior, where the threshold displays a non-linear relationship with α . When present, Segment 4 returns to linear decline, but the threshold maintains a non-zero value even when α approaches 0.5. Depending on the information structure, certain segments may be absent, and this boundary presents as two or three segments.

Combining the two boundaries yields a key insight: higher coordination ability produces a more selective set of stable outcomes.

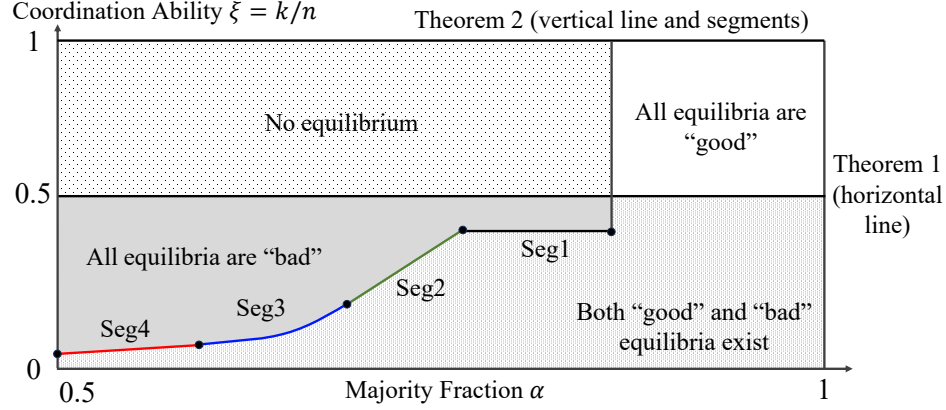


Figure 1: Illustration of boundaries characterized in Theorem 1 and Theorem 2 on the coordination ability and the majority fraction. “Good” and “bad” refer to equilibria that reach the informed decision and those that do not reach the informed decision, respectively.

When coordination ability is high, equilibria either always reach informed decisions (if preferences are sufficiently aligned) or fail to exist entirely (if preferences are divergent). Low coordination ability, conversely, tolerates both informed and uninformed equilibria, leading to mixed predictions about outcomes.

Moreover, we construct explicitly the k -equilibrium reaching the informed majority decision with the largest possible k for each segment (Section E). Like the segmental boundary curve, equilibria under different α exhibit different structures, revealing the non-trivial behavior of strategic agents with different coordination ability. For example, when the majority fraction is large (in Segment 1 and 2), all agents whose preferences are the minority converge on the same alternative in an equilibrium. However, as this fraction approaches $\frac{1}{2}$ (in Segment 4), minority agents split evenly between two alternatives in the equilibrium.

We also analyze the relationship between the shape of the boundaries and the distribution of the private information of the agents (Section 4.3). When agents possess more accurate private information, the threshold ξ^* value increases. This observation aligns with the intuition that agents with more precise information are more certain about their preferences, thereby increasing the likelihood of reaching a stable outcome.

Technical Overview. At a high level, the existence of an equilibrium hinges on the fraction of votes each alternative gets in expectation (Equation 1 and 2). For an equilibrium to exist, the informed majority decision must secure more than half of the expected votes under both ground truth states, even when a fraction $\frac{k}{n}$ of agents deviates from their prescribed strategies. Consequently, the majority agents employ strategies that maximize their margin above the $\frac{1}{2}$ threshold in expected vote share. Minority agents, conversely, work to diminish this margin and push the vote share below the majority threshold. The critical deviation threshold is determined by three competing factors: a majority’s ability to build its margin, its requirement to maintain the margin under both ground truth states, and a minority’s ability to deviate. This interplay makes finding the threshold a complex analytical challenge.

2 Related Works

The study of binary voting with incomplete information traces back to the famous Condorcet Jury Theorem [13], which has “formed the basis for the development of social choice and collective decision-making as modern research fields” [45]. The Condorcet Jury Theorem considers the scenario where agents honestly reflect their information in the vote and gives merit to the majority vote to make a correct decision. Extensive literature generalize the Condorcet Jury Theorem into a wider range of scenarios, including correlated private information [44, 49], agents with different competencies [9, 43], and plurality voting with more alternatives [26, 52]. Experimental studies have also been conducted to examine the Theorem in real-world scenarios [8, 19, 25]. Austen-Smith and Banks [6] pioneer the study of the Condorcet Jury Theorem under a game theory context and show that the classical non-strategic behaviors may fail to form a Nash equilibrium. Consequently, a large literature focuses on the strategic behaviors of the voters and the existence of an equilibrium that reaches a correct outcome [21, 22, 39, 40, 42, 50].

A recent line of work brings up the coalitional strategic behavior in voting with incomplete information. Schoenebeck and Tao [48] design a mechanism to incentivize truthful reporting and reach the outcome that favors the majority as the ground truth is known to all. Han et al. [29] reveals a surprising equivalence between strategies reaching the informed majority decision and those forming an (approximated) strong equilibrium. Their work provides strong support for the majority vote reaching a good outcome via strategic behavior. Both studies adopt a similar model where agents have diverse preferences yet retain some common value. Deng et al. [15] study agents with drastically different preferences and characterize conditions under which a strong equilibrium exists. All three papers apply the solution concept of ε -strong Bayes Nash equilibrium in which no group (with unlimited size) of agents has incentives to deviate.

While Austen-Smith and Banks [6] and many works [14, 17, 50] assume identical utility functions across agents, a subsequent research has explored more diverse utility structures. In the first

direction, agents have different ordinal preference yet share some common values [21, 24, 29, 41, 48]. These models typically feature three agent types: two groups with fixed preferences for two alternatives respectively, and a third group whose preferences align with the ground truth. In the second direction, which our paper follows, agents have opposite preferences contingent on the ground truth [3, 10, 11, 37].

Our study also closely relates to research on group strategic behavior, particularly those with limited capacity. There is extensive literature [7, 16, 31, 33, 46, 51] examining group-strategic behavior, which originated from the concept of strong Nash equilibrium in complete information games [5]. In the context of Bayesian games, group strategic behavior has been investigated across multiple domains, including cooperative games [34, 35], implementation theory in mechanism design [28, 47], and voting. Recent research has highlighted the importance of examining capacitated coalitional behavior and has introduced various solution concepts for this scenario. Guo and Yannelis [27] propose coalitional interim equilibrium parameterized by an exogenously given set of all admissible coalitions. Han et al. [30] propose (ex-ante) Bayesian k -strong equilibrium parameterized by the largest capacity of the deviating group k . Abraham et al. [2] also proposes a k -equilibrium where the deviators will share all their private information in the group. Abraham et al. [1] study capacitated strategic behavior under a non-Bayesian game. The introduction of capacitated strong equilibria enables theories to model a broader range of real-world scenarios and provides more nuanced insights, particularly in situations where unlimited strong equilibria are too restrictive to exist [23].

3 Preliminaries

n agents vote for two candidates **A** (accept) and **R** (reject). The majority rule is applied, and **A** becomes the winner if and only if it gets at least $\frac{n}{2}$ votes. There is a world state (ground truth) $W \in \{L, H\}$ (Low and High) that affects the preferences of the agents. The common prior on the world state is denoted by P_H and P_L , where $P_H + P_L = 1$. The world state is not directly observable. Instead, each agent receives a private signal $m \in \{\ell, h\}$ whose distribution $\mathbf{P} = (P_{mW})_{m \in \{\ell, h\}, W \in \{L, H\}}$ depends on the world states, representing the private information each agents possess. The signals of different agents are i.i.d. conditioned on the world state. For a world state W and a signal m , let P_{mW} denote the probability that an agent receives m signal when the world state is W . We assume that an h signal is more likely to occur in world state H than in world state L , i.e., $P_{hH} > P_{hL}$, and vice versa. Let $\Delta = P_{hH} - P_{hL}$ be the difference on the signal distribution under two world states. From the fact that $P_{hH} + P_{\ell H} = P_{\ell L} + P_{hL} = 1$, we have the following observation.

PROPOSITION 1. $\Delta = P_{hH} - P_{hL} = P_{\ell L} - P_{\ell H} = P_{hH} \cdot P_{\ell L} - P_{hL} \cdot P_{\ell H} = P_{hH} + P_{\ell L} - 1$.

For each agent i , the utility function $v_i : \{H, L\} \times \{\mathbf{A}, \mathbf{R}\} \rightarrow \{0, 1, \dots, B\}$ represents the preference of the agent. In this setting, we consider two types of agents. The majority type prefer **A** in world state H ($v(H, \mathbf{A}) > v(H, \mathbf{R})$) and **R** in world state L ($v(L, \mathbf{A}) < v(L, \mathbf{R})$), and the minority type prefer **R** in world state H ($v(H, \mathbf{A}) < v(H, \mathbf{R})$) and **A** in world state L ($v(L, \mathbf{A}) > v(L, \mathbf{R})$). Let $\alpha \in (0.5, 1]$ be the (approximated) fraction of the majority agents, where the number of majority agents $n_{maj} = \lfloor \alpha \cdot n \rfloor$. We define the

approximation α because our study is asymptotic in the number of agents n .

The *informed majority decision* is the decision to be made if all the agents knew the world state. In this setting, the informed majority decision follows the majority type agents' preference: the informed majority decision is **A** in world state H and **R** in world state L .

EXAMPLE 2 (INFORMATION STRUCTURE AND AGENT PREFERENCES). Consider the legislator voting scenario in Example 1. Suppose $n = 50$ agents vote on the economic policy. The world state is the outcome of the project, which can be a high inflation rate (H) or a low inflation rate (L). The prior that a high rate occurs is $P_H = 0.6$ and $P_L = 0.4$. The distribution is of private signals $P_{hH} = 0.7$, $P_{\ell H} = 0.3$, $P_{hL} = 0.2$, and $P_{\ell L} = 0.8$. This means that an agent receives an h signal with probability 0.7 when the world state is H and probability 0.2 when the state is L . The utilities of the agents are illustrated in Table 1. In this example, we assume that agents of the same type share an identical utility function, which may not necessarily be true in general.

Table 1: Utility of agents.

Type	$v_i(H, \mathbf{A})$	$v_i(L, \mathbf{A})$	$v_i(H, \mathbf{R})$	$v_i(L, \mathbf{R})$
Majority (Doves)	4	0	1	2
Minority (Hawks)	2	3	3	1

A strategy σ of an agent i maps his/her signal to a distribution on $\{\mathbf{A}, \mathbf{R}\}$, denoting the action of the agent. A strategy profile Σ is a vector of the strategies of all the agents. We also represent a strategy σ_i in the following way: for an agent i , $\sigma_i = (\beta_h^i, \beta_\ell^i)$, where β_h^i and β_ℓ^i are the probability that i votes for **A** when his/her private signal is ℓ and h , respectively.

The *fidelity* of a strategy profile is the likelihood that the informed majority decision is reached under this strategy profile. Let $\lambda_W^{\mathbf{A}}(\Sigma)$ and $\lambda_W^{\mathbf{R}}(\Sigma)$ be the probability that **A** (**R**, respectively) becomes the winner when the world state is W and the strategy profile is Σ . Then the fidelity has the following form:

$$A(\Sigma) = P_H \cdot \lambda_H^{\mathbf{A}}(\Sigma) + P_L \cdot \lambda_L^{\mathbf{R}}(\Sigma).$$

Similarly, we can define the (ex-ante, before observing his/her signal) expected utility of an agent i under a strategy profile Σ :

$$u_i(\Sigma) = P_L(\lambda_L^{\mathbf{A}}(\Sigma) \cdot v_i(L, \mathbf{A}) + \lambda_L^{\mathbf{R}}(\Sigma) \cdot v_i(L, \mathbf{R})) \\ + P_H(\lambda_H^{\mathbf{A}}(\Sigma) \cdot v_i(H, \mathbf{A}) + \lambda_H^{\mathbf{R}}(\Sigma) \cdot v_i(H, \mathbf{R})).$$

Non-dominated strategies. Our theoretical results concern equilibria in which agents employ only non-dominated strategies. A strategy is dominated if playing a different strategy always brings the agent a higher expected utility.

Definition 1. A strategy σ'_i is an (ex-ante) weakly dominated strategy for agent i if there exists a strategy σ_i such that, regardless of what other agents play Σ_{-i} , $u_i((\sigma'_i, \Sigma_{-i})) \leq u_i((\sigma_i, \Sigma_{-i}))$, and there exists a Σ_{-i} such that $u_i((\sigma'_i, \Sigma_{-i})) < u_i((\sigma_i, \Sigma_{-i}))$.

The constraint of no agents playing weakly dominated strategies follows the convention in the literature [20, 21, 50], and we believe this restriction is mild and reasonable. Firstly, an agent playing a dominated strategy has incentives to switch to a non-dominated strategy unilaterally, which brings no utility loss in any

circumstances and utility gain in some circumstances. This contradicts the intention of an equilibrium that predicts a stable outcome where strategic agents aim to maximize their utilities. Secondly, as shown in the following lemma, whether a strategy is (weakly) dominated can be easily determined, and playing a non-dominated strategy does not bring extra computational cost and difficulties to the agents.

LEMMA 1. *For an arbitrary agent i , a strategy $\sigma = (\beta_\ell, \beta_h)$ is NOT a weakly dominated strategy if and only if the following conditions hold.*

- If i is a majority agent: $\beta_\ell = 0$ or $\beta_h = 1$.
- If i is a minority agent: $\beta_\ell = 1$ or $\beta_h = 0$.

The idea to prove Lemma 1 is in Appendix B.

Sequence of Environments. An environment $\mathcal{I}$ contains the agent number n , the world state prior distribution (P_L, P_H) , the signal distributions $\mathbf{P}$, the utility functions $\{v_i\}_{i=1}^n$, and the approximated fraction of majority agents α . Let $\{\mathcal{I}_n\}_{n=1}^\infty$ (or $\{\mathcal{I}_n\}$ for short) be a sequence of environments, where each $\mathcal{I}_n$ is an environment with n agents. The environments in a sequence share the same prior distributions, signal distributions, and approximated fractions of each type. There are no additional assumptions on the utility functions. A sequence of strategy profiles $\{\Sigma_n\}$ is defined on a sequence of environment $\{\mathcal{I}_n\}$, where for each n , Σ_n is a strategy profile in $\mathcal{I}_n$.

A strategy profile $\Sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ is an ε -ex-ante Bayesian k -strong equilibrium $((\varepsilon, k)$ -EBSE) if there does not exist a subset of agents D (the *deviating group*) with $|D| \leq k$ and a strategy profile $\Sigma' = (\sigma'_1, \sigma'_2, \dots, \sigma'_n)$ (the *deviating strategy profile*) such that

- (1) $\sigma_i = \sigma'_i$ for all $i \notin D$;
- (2) $u_i(\Sigma') \geq u_i(\Sigma)$ for all $i \in D$; and
- (3) there exists $i \in D$ such that $u_i(\Sigma') > u_i(\Sigma) + \varepsilon$.

By definition, when $\varepsilon = 0$, the equilibrium is an ex-ante Bayesian k -strong equilibrium where no group of at most k agents can strictly increase their utilities through deviation; and when $k = n$, the equilibrium is an ε -ex-ante Bayesian strong equilibrium (named ε -strong Bayes-Nash equilibrium in [15, 29, 48]). In this paper, we name k the coordination ability of agents, and its fraction version is $\xi = \frac{k}{n}$.

4 Main Results: Equilibrium Analysis

The main results of this paper are summarized in Figure 1, which partitions sequences of environments by (1) whether all sequences of equilibria reach the informed majority decision and (2) whether there exists a sequence of equilibria that reaches the informed majority decision (a “good” equilibrium). The partition depends on the signal distribution $\mathbf{P}$, majority fraction α , and the coordination ability fraction ξ . Section 4.1 characterizes the boundary determining when a “bad” equilibrium exists (so that not all equilibria are “good”), and Section 4.2 characterizes the boundary determining when at least one “good” equilibrium exists. Combining the two boundaries yields our full characterization in Figure 1, implying that higher coordination ability produces more selective stable outcomes, while low coordination ability leads to more tolerated, mixed outcomes. In Section 4.3, we discuss how signal distribution $\mathbf{P}$ affects the boundaries.

We formalize the definition of a “good” or “bad” equilibrium, named as *informed* and *uninformed* equilibrium, respectively.

Definition 2. A sequence of profiles $\{\Sigma_n\}$ is an *informed* ξ -strong equilibrium if it satisfies the following conditions.

- (1) for all n , there is no weakly dominated strategies in Σ_n ,
- (2) the fidelity $A(\Sigma_n)$ converges to 1, and
- (3) there exists a sequence $\{\varepsilon_n\}$ such that ε_n converges to 0 and that for all n , Σ_n is an ε_n -ex-ante Bayesian ξn -strong equilibrium.

A sequence of profiles $\{\Sigma_n\}$ is an *uninformed* ξ -strong equilibrium if it satisfies condition (1) and (3), but the fidelity $A(\Sigma_n)$ does not converge to 1. Specifically, $\xi = 1$ implies the sequence is asymptotically an ε_n -ex-ante Bayesian strong equilibrium.

4.1 Worst Case Equilibrium

We first characterize the boundary deciding when all k -strong equilibria reach the informed majority decision. The answer to this question is affirmative if and only if $\xi > \frac{1}{2}$ ($k > \frac{n}{2}$). Below this threshold, there always exists at least one uninformed ξ -strong equilibrium.

THEOREM 1. *For any $\xi \in [0, 1]$ and any sequence of instance $\{\mathcal{I}_n\}$, an uninformed ξ -strong equilibrium exists if and only if $\xi \leq \frac{1}{2}$.*

Theorem 1 contributes to one of the two partitions in Figure 1 (the horizontal $\frac{k}{n} = 0.5$). When $\xi \leq \frac{1}{2}$, the existence of an uninformed equilibrium implies that “all equilibria are good” does not hold. When $\xi > \frac{1}{2}$, once there exists a k -strong equilibrium, it reaches the informed majority decision. At a high level, this observation connects to the Price of Anarchy [38], which measures the ratio between optimal social welfare and the worst equilibrium outcome. Using fidelity as a proxy for social welfare, Theorem 1 implies that the Price of Anarchy approaches one only when society possesses high coordination ability.

4.2 Best Case Equilibrium

We then characterize the boundary deciding when there exists a k -strong equilibrium reaching the informed majority decision. The boundary depends on the signal distribution $\mathbf{P}$, the majority fraction α , and the coordination ability fraction ξ , and can be represented by two threshold functions θ and ξ^* . The first threshold θ takes the signal distribution $\mathbf{P}$ as input and outputs a value in $(\frac{1}{2}, 1)$. When the majority fraction α exceeds $\theta(\mathbf{P})$, an informed 1-strong equilibrium (a sequence of “good” n -strong equilibria) exists. The second threshold ξ^* takes both $\mathbf{P}$ and α as inputs and outputs a value in $(0, \frac{1}{2})$. When $\alpha \leq \theta(\mathbf{P})$, an informed ξ -equilibrium exists if and only if $\xi < \xi^*(\mathbf{P}, \alpha)$.

THEOREM 2. *There exist threshold functions θ and ξ^* such that for any sequence of instance $\{\mathcal{I}_n\}$ with signal distribution $\mathbf{P}$ and majority fraction α ,*

- (1) if $\alpha > \theta(\mathbf{P})$, an informed 1-strong equilibrium exists.
- (2) if $\frac{1}{2} < \alpha \leq \theta(\mathbf{P})$, an informed ξ -strong equilibrium exists if and only if $\xi < \xi^*(\mathbf{P}, \alpha)$.

Theorem 2 contributes to the other partition in Figure 1, yet it is much more complicated. In the order of α decreasing, this partition starts with a critical point θ where the largest ξ that tolerates

an informed equilibrium declines drastically. The boundary then follows with at most four segments, depending on the signal distribution. These segments, ordered from highest to lowest values of α as Segment 1 to 4, are given descriptive names based on their distinct features. The “Flat” Segment 1 begins at θ and maintains a constant value ξ^* as α decreases. In the “Steep” Segment 2, ξ^* decreases linearly with α . Segment 3, the “Non-Linear” segment, is characterized by a non-linear relationship between ξ^* and α . When the signal distribution is inaccurate or symmetric (satisfying $(|P_{tL} - P_{hH}| < \min(P_{hL}, P_{tH}))$), Segment 3 ends before $\alpha = \frac{1}{2}$, there will be a “Tail” Segment 4; otherwise, the boundary ends with three segments. We defer the explicit form of θ and ξ^* (which is very complicated) to Appendix A.

A special case when the signal is symmetric, i.e., $P_{hH} = P_{tL}$. Under this specific setting, both the “Flat” Segment 1 and the “Non-Linear” Segment 3 reduce to points, and ξ^* consists of only “Steep” Segment 2 and “Tail” Segment 4 as shown in Figure 2(a).

EXAMPLE 3. Here is an example adopting the scenario in Example 2 to illustrate the thresholds θ and ξ^* in Theorem 2. Note that $P_{hH} = 0.7$ and $P_{tL} = 0.8$ imply that this instance has all four segments. In this case, the first threshold $\theta = \frac{11}{16}$. For the threshold in the non-linear segment, we give a range of $\Theta(\frac{1}{\alpha})$ rather than the explicit form.

Table 2: Range and Threshold for each segment.

Segment	Range of α	Threshold ξ^*
$\alpha > \theta$	$(\frac{11}{16}, 1)$	Unbounded
Flat 1	$[\frac{5}{8}, \frac{11}{16}]$	$\frac{5}{16}$
Steep 2	$[0.556, \frac{5}{8})$	$\frac{5}{2}(\alpha - \frac{1}{2})$
Non-linear 3	$[\frac{10}{19}, 0.556)$	$\Theta(\frac{1}{\alpha})$
Tail 4	$(\frac{1}{2}, \frac{10}{19})$	$\frac{1}{4}\alpha$

Likewise, Theorem 2 connects to the Price of Stability [4], which measures the ratio between optimal social welfare and the best-case equilibrium outcome. Using fidelity as a proxy for social welfare, Theorem 2 implies that the Price of Stability approaches one if and only if society’s coordination ability is below the threshold.

4.3 Shape Analysis on the Boundary

In this section, we discuss how the signal distribution $\mathbf{P}$ affects the shape of the boundary characterized in Theorem 2.

Accuracy. As the signals become more accurate (i.e., P_{hH} and P_{tL} increase), two effects emerge in the equilibrium threshold ξ^* . First, the interval $(\frac{1}{2}, \theta]$, where ξ^* takes effect, contracts because θ is negatively correlated with signal accuracy. Second, the equilibrium threshold ξ^* increases within each segment of its piecewise function. This observation aligns with intuition: as agents receive more precise signals about the true state, the majority group can more effectively leverage their information advantage to secure their preferred outcome.

Asymmetry. As the signal asymmetry $P_{tL} - P_{hH}$ increases, there are corresponding changes in the intervals of Segments 1 and 4. The length of Flat Segment 1 exhibits a positive linear relationship with the signal asymmetry. Conversely, “Tail” Segment 4 contracts as the signal asymmetry increases, since its left boundary is a decreasing function of $(P_{tL} - P_{hH})$ while its right boundary remains fixed at $\frac{1}{2}$.

Non-vanishing. Regardless of the signal distribution, ξ^* does not converges to 0 when α goes to $\frac{1}{2}$. This indicates that, even when the majority only slightly outnumbers the minority, an unorganized minority still yields a stable outcome favoring the 51%.

Figure 2 illustrates the shapes of ξ^* under different distributions. For example, Figure 2(a) with a symmetric signal distribution contains only Segment 2 and 4, while Figure 2(b) with an asymmetric signal distribution has a much larger “Flat” Segment 1 and shorter Segment 4. In Figure 2(c) where the signal is inaccurate, the interval of ξ^* is long yet the value is close to 0.

5 Proof Sketch

Section 5.1 introduces our key analytical tool, the expected vote share framework, which connects expected utility, fidelity, and strategic behavior through a max-Min perspective on equilibrium analysis. Section 5.2 applies this framework to prove Theorem 2, transforming the problem into a tractable optimization. Section E extends the proof sketch by exhibiting the equilibria achieving the threshold ξ^* across different segments, each displaying distinct strategic behaviors. Finally, Section 5.3 provides the proof sketch for Theorem 1.

5.1 Key Framework: Expected Vote Share.

Given a strategy profile Σ , its expected vote share is the (ex-ante) expected fraction of votes for **A** conditioned on two world states, respectively. Specifically, when each agent i ’s strategy is $\sigma_i = (\beta_h^i, \beta_t^i)$, the expected vote share has the following form.

$$f_H = \frac{1}{n} \cdot \sum_{i=1}^n (P_{hH} \cdot \beta_h^i + P_{tH} \cdot \beta_t^i). \quad (1)$$

$$f_L = \frac{1}{n} \cdot \sum_{i=1}^n (P_{hL} \cdot \beta_h^i + P_{tL} \cdot \beta_t^i). \quad (2)$$

Han et al. [29] reveals the asymptotical relationship between the expected vote share and the fidelity for a sequence of profiles: with a few exceptions, the fidelity converges to 1 if and only if $f_H > \frac{1}{2}$ and $f_L < \frac{1}{2}$ holds for all sufficiently large n . At a high level, this is because agents’ votes can be viewed as independent Bernoulli variables, and their sum will concentrate around the expectation with high probability as n increases. A more formal statement is in Appendix C.

Now consider a sequence of strategy profiles $\{\Sigma_n\}$ that satisfies (1) no agents play weakly dominated strategies and (2) the fidelity converges to 1. We establish that only the minority agents need to be considered in potential deviation. This is because any sequence of deviating strategy profiles falls into one of two cases: either its fidelity still converges to 1, making utility changes negligible for all agents, or its fidelity fails to converge to 1, in which case majority agents experience strictly lower utility and have no incentive to deviate.

Consequently, we examine only deviations by minority agents, whose utilities are negatively correlated with fidelity. By the relationship between the fidelity and the expected vote share, a group of minority agents could successfully deviate if and only if they could create a new strategy profile (with expected vote share f'_H

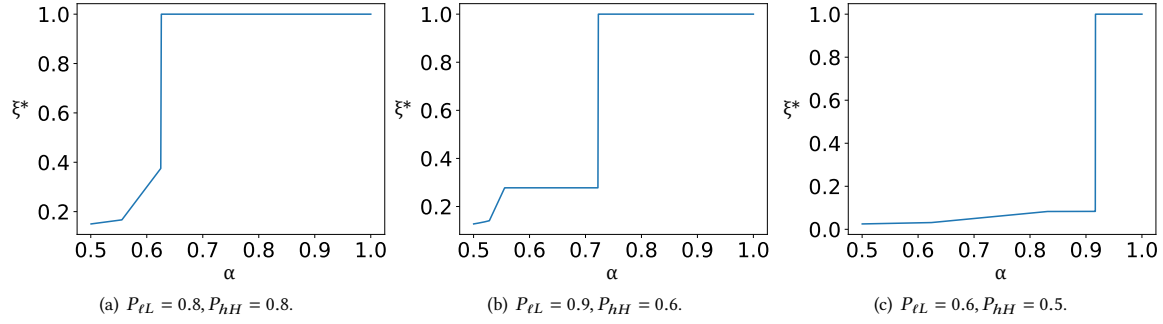


Figure 2: The shape of ξ^* under different signal distribution. We set $\xi^* = 1$ when $\alpha > \theta$, indicating the existence of a strong equilibrium.

and f'_L) such that either

$$f'_H \leq \frac{1}{2}, \quad (3)$$

leading to the expected majority being **R** in state *H*, or

$$f'_L \geq \frac{1}{2}, \quad (4)$$

leading to the expected majority being **A** in state *L*.

At a high level, characterizing the boundary can be reinterpreted as finding the boundary of a maximin game. First, the Max player, aiming to maintain the expected vote share from the border $\frac{1}{2}$, chooses a sequence of profiles $\{\Sigma_n\}$. Then, the Min player, aiming to break the border, chooses a group of $\xi \cdot n$ deviators and their strategies. ξ^* is the smallest ξ such that the Min player wins, and no strategy profiles can tolerate a group of $\xi \cdot n$ deviators and maintain the expected vote share.

For the Min player, the optimal action is to embrace one of two extremes. In the first extreme, the goal is to maximize f'_L , the expected fraction of **A** votes in world *L* after deviation. The deviators switch to always vote for **A**, and consist of the minority agents who are initially least likely to vote for **A**. In the second extreme, the goal is to minimize f'_H , the expected fraction of **A** votes in world *H* after deviation. Here, the deviators switch to always vote for **R**, and consist of the minority agents who are initially least likely to vote for **R**. If either deviation succeeds, they succeed; and if both fail, then no deviation will succeed, implying the original strategy profile is an equilibrium.

For the Max player, the optimal action is non-trivial. In general, three correlated factors may affect how large a deviating group, ξ , a particular strategy profile can tolerate.

- The first factor is the difference between f_H and f_L , which is determined by the quantity $\beta_h^i - \beta_\ell^i$ of each agent *i*. When $f_H - f_L$ is large, it is at least possible for the threshold of $1/2$ to be between the two values and not too close to either of them.
- Secondly, the strategies of the minority agents. Agents' ability to change f_H and f_L depends on their initial strategy. For example, an agent who always votes for **A** cannot help to increase f_L but can help to increase f_H by $\frac{1}{n}$. Meanwhile, an agent who votes for **A** with signal ℓ and **R** with signal h can deviate to either increase f_L by $\frac{P_{hL}}{n}$ or to decrease f_H by $\frac{P_{\ell H}}{n}$.

- The last factor is balance. Because the deviators have two possible directions to deviate (decrease f_H and increase f_L), a deviation-proof strategy profile needs to keep both margins $f_H - \frac{1}{2}$ and $\frac{1}{2} - f_L$ sufficiently large to overwhelm the aforementioned deviating power of minority agents. Sometimes deviations can be neutralized by keeping margin balanced even at the expense of decreasing the difference $f_H - f_L$.

The interplay of these three factors largely contributes to the complexity of the problem.

5.2 Proof Sketch of Theorem 2

In the case where $\alpha > \theta$, the proof follows that of Theorem 3.1 in [15], where they show the existence of a (sequence of) ε -strong Bayes-Nash equilibrium, where ε converges to 0, reaching the informed majority decision.

For the case where $\alpha \leq \theta$, our analysis applies the expected vote-share framework: a sequence of strategy profiles is an informed ξ -equilibrium if and only if the expected vote share margin $f'_H - \frac{1}{2}$ and $\frac{1}{2} - f'_L$ remain strictly positive even under the two extreme deviations by ξ fraction of minority agents. Therefore, to establish the lower bound, we find the maximum ξ such that these margin constraints hold, along with the strategy profile that achieves this maximum. For the upper bound, we show that no strategy profile can satisfy the constraints for any larger ξ .

The first challenge is that the search space for optimal strategy profiles encompasses all strategy profiles that reach informed majority decisions, a vast and complex space where the relationship between strategies and the optimum is highly nontrivial. To make the problem tractable, our first step is to reduce this search space to a finite set of parameters. First, the majority agents need not be analyzed individually. Since the majority agents never deviate in equilibrium, their impact on expected vote shares depends only on their average strategy β_h^{maj} and β_ℓ^{maj} .

Minority agents require more careful treatment, as strategies with identical average strategies can yield different deviation capabilities. We exploit the structure of non-dominated strategies, which satisfy either $\beta_h = 0$ or $\beta_\ell = 1$. This allows us to partition minority agents into two groups: Group 0 plays strategies with $\beta_h = 0$, and Group 1 plays strategies with $\beta_\ell = 1$. Since our goal is to find the maximum sustainable ξ , we can focus on strategy

profiles that minimize minority deviation power. We establish that this minimum is achieved when all agents within each group play identical strategies. Specifically, Group 0 agents play symmetrically ($\beta_h^0 = 0, \beta_\ell^0$), and Group 1 agents play symmetrically ($\beta_h^1, \beta_\ell^1 = 1$). Consequently, these reductions yield a five-parameter characterization of the strategy search space: the majority's average strategy β_h^{maj} and β_ℓ^{maj} , Group 0's strategy β_ℓ^0 , Group 1's strategy β_h^1 , and the fraction of Group 0, denoted by γ . This substantial reduction enables transforming the problem into a tractable optimization problem.

Formulating the Optimization Problem. With a strategy profile characterized by these parameters, the two extreme deviation patterns become explicit. In the first, all deviators come from Group 0 and switch to always vote for A, maximizing f'_L and minimizing the L-side margin. In the second, all deviators come from Group 1 and switch to always vote for R, minimizing f'_H and thus minimizing the H-side margin. (Cases where the deviating group exceeds either group size are handled separately as corner cases.) Consequently, the margin constraints can be written where f'_H and f'_L are functions of five parameters, α and ξ .

$$\begin{aligned} f'_L(\beta_h^{maj}, \beta_\ell^{maj}, \beta_\ell^0, \beta_h^1, \gamma, \alpha, \xi) &< \frac{1}{2}. \\ f'_H(\beta_h^{maj}, \beta_\ell^{maj}, \beta_\ell^0, \beta_h^1, \gamma, \alpha, \xi) &> \frac{1}{2}. \end{aligned}$$

Combining these constraints with the feasibility requirement $(\beta_h^{maj}, \beta_\ell^{maj}) \in [0, 1]^2$, we eliminate the majority strategy parameters and reformulate the constraints where ξ is isolated:

$$\begin{aligned} \xi &< \xi_h(\beta_\ell^0, \beta_h^1, \gamma, \alpha), \\ \xi &< \xi_\ell(\beta_\ell^0, \beta_h^1, \gamma, \alpha). \end{aligned}$$

For each α , the threshold $\xi^* = \max_{\beta_\ell^0, \beta_h^1, \gamma} \min(\xi_h, \xi_\ell)$. This formulation has a clear interpretation: for any $\xi < \xi^*$, the parameters maximizing $\min(\xi_h, \xi_\ell)$ constitute an optimal strategy profile satisfying the margin constraints and lead to an informed ξ -strong equilibrium; for $\xi \geq \xi^*$, no parameter choice can satisfy the constraints, so no equilibrium exists. The remaining analysis reduces to solving this non-convex optimization problem, where different values of α lead to different optima and parameter configurations (thus different equilibria) that achieve them. The full proof is in the full version of this paper, and a description of equilibria in each segment is in Appendix E.

5.3 Proof Sketch for Theorem 1.

For $\xi \leq \frac{1}{2}$, there is an easy proof for the existence of a k -strong equilibrium not reaching the informed majority decision: the strategy profile where all the agents unanimously vote for A. Under this strategy profile, A is elected the winner even under world state L, yet no group of size at most $\frac{n}{2}$ agents can flip the outcome (note that A is favored by the tie-breaking).

For $\xi > \frac{1}{2}$, the proof follows the same high-level approach of leveraging expected vote share, but now the deviators are majority agents. With a deviating coalition exceeding half the population, majority agents can construct deviating strategies that reach the informed majority decision. The construction proceeds as follows: one fraction of deviators plays strategies that "complement" the non-deviators, setting the expected vote shares f'_H and f'_L both

slightly below $\frac{1}{2}$. The remaining fraction votes informatively to maximize $f'_H - f'_L$, ensuring f'_H exceeds $\frac{1}{2}$ while f'_L remains below it. This deviating strategy achieves fidelity converging to 1, providing majority agents with non-negligible benefits from deviation. The complete proof appears in Appendix D.

6 Conclusion and Future Works

We study a voting problem with two opposed groups whose preferences depend on a ground truth that cannot be directly observed. Conflicting preferences, incomplete information, and group strategic behavior pose challenges to the problem. Previous analysis based on the Bayesian Nash equilibrium or strong equilibrium is either overly inclusive or exclusive. We adopt the solution concept of ε -approximate ex-ante Bayesian k -strong equilibrium, in which no group of at most k agents has incentives to deviate, to study scenarios where agents are less organized and have limited power to coordinate on strategic behavior. We give closed-form boundaries on (1) when all equilibria are informed and (2) when there exists an informed equilibrium. The boundary includes multiple drastically different segments, including a highly non-trivial non-linear segment. Our results provide a finer prediction of the outcome of strategic behavior in the majority vote.

Several natural extensions serve as interesting future directions for our work. Although we fully characterize the existence of k -strong equilibrium in favor of the majorities, the question remains open regarding the equilibrium in favor of the minorities, which will be an extension of the current ξ^* to the range $\frac{1}{2} > \alpha > 0$. The second direction is to consider a full spectrum of preferences among agents, including those who always prefer one alternative and those whose preferences depend on the world state. Finally, it is also interesting to study the problem of a more structured agent relationship. For example, the relationship of agents is represented as edges on a graph, and a group will deviate only if they form a clique or have enough edges.

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A Explicit Form of the Thresholds.

In this section, we give the explicit form of the thresholds θ and ξ^* .

$$\theta = \begin{cases} \frac{1}{2} + \frac{P_{tH}}{2P_{tL}} & P_{tL} \geq P_{hH}, \\ \frac{1}{2} + \frac{P_{hL}}{2P_{hH}} & P_{tL} < P_{hH}. \end{cases}$$

For ξ^* , assume without loss of generality that $P_{hH} \leq P_{tL}$. In this case, $\theta = \frac{1}{2} + \frac{P_{tH}}{2P_{tL}}$. Given a fixed information structure $\mathbf{P}$, we explicitly give the formula of ξ^* as a function of α . We will also abuse this notation of $\xi^*(\alpha)$ in the rest of the appendix.

$$\begin{array}{l|l} \text{When } \frac{1}{1+P_{tL}+(P_{tL}-P_{hH})} > \frac{1}{2}, & \text{When } \frac{1}{1+P_{tL}+(P_{tL}-P_{hH})} \leq \frac{1}{2}, \\ \xi^*(\alpha) = \begin{cases} \frac{\frac{\Delta}{2P_{tL}}}{\frac{\Delta \cdot (\alpha-1/2)}{P_{hL}}} & \theta \geq \alpha \geq \frac{1}{2P_{tL}} \\ \frac{1}{2P_{tL}} & \frac{1}{2P_{tL}} > \alpha \geq \alpha_{NL} \\ \xi_{NL} & \alpha_{NL} > \alpha > \frac{1}{1+P_{tL}+(P_{tL}-P_{hH})} \\ \frac{1}{2} \cdot \Delta \cdot \alpha & \frac{1}{1+P_{tL}+(P_{tL}-P_{hH})} > \alpha > \frac{1}{2} \end{cases} & \xi^*(\alpha) = \begin{cases} \frac{\frac{\Delta}{2P_{tL}}}{\frac{\Delta \cdot (\alpha-1/2)}{P_{hL}}} & \theta \geq \alpha \geq \frac{1}{2P_{tL}} \\ \frac{1}{2P_{tL}} & \frac{1}{2P_{tL}} > \alpha \geq \alpha_{NL} \\ \xi_{NL} & \alpha_{NL} > \alpha > \frac{1}{2} \end{cases} \end{array}$$

Here, ξ_{NL} is a non-linear function on α which makes up the “Non-Linear” Segment 3 in the curve. α_{NL} is the changing point between Segment 2 and 3 and comes from solving the inequality $\xi_{NL} \geq \frac{\Delta(\alpha-1/2)}{P_{hL}}$.

$$\alpha_{NL} = \frac{2P_{hH}P_{tL}^2 + P_{hH} \cdot (P_{hH} + 3P_{tL}) - (3P_{hH} + P_{tL}) + 2 - 2P_{hL} \cdot \sqrt{P_{hH}P_{hL}P_{tH}P_{tL}}}{2P_{tH}^2 + 8P_{hH}P_{tL}^2}. \quad (5)$$

$$\xi_{NL} = \frac{\Delta}{4\sqrt{2(1-\alpha P_{tH})(1-2\alpha P_{tL})P_{tH}P_{tL}} + 2P_{tH} + 4P_{tL} - 8\alpha P_{tH}P_{tL}}. \quad (6)$$

It is not hard to see that $\theta > \frac{1}{2}$. For ξ^* , Note that $\Delta = P_{tL} - P_{tH} < P_{tL}$, so the $\xi^* < \frac{1}{2}$ in Segment 1. Given the decreasing nature of ξ^* , it's smaller than $\frac{1}{2}$ for all $\alpha \in (\frac{1}{2}, \theta]$.

B Proof Sketch of Lemma 1

We give the reasoning for a majority agent, and that for a minority agent is similar. A majority agent's expected utility increases with the probability that **A** wins in world state H and decreases with the probability that **R** wins in world state H . Given that a signal h implies the world state H is more likely than a signal l , increasing β_h and decreasing β_l usually helps the agent to increase his expected utility. Therefore, a strategy where $\beta_l > 0$ and $\beta_h < 1$ is dominated by a strategy with a larger β_h and a smaller β_l . On the other hand, when $\beta_l = 0$ or $\beta_h = 1$, any change in the strategy decreases the expected utility under world state H or under world state L . Therefore, there will exist a strategy profile of all other agents where the loss exceeds the gain in expectation, and the expected utility of the agent decreases. Therefore, in this case, we can show that the strategy profile is not weakly dominated.

C Expected Vote share and fidelity.

[29] provides useful tool to determine the fidelity of a profile sequence by the *expected vote share* of alternative **A** under different world states.

Definition 3 (Expected vote share). Given an instance of n agents, and a strategy profile Σ , let random variable X_i^n be “agent i votes for **A**”: $X_i^n = 1$ if agent i votes for **A**, and $X_i^n = 0$ if i votes for **R**. Then the expected vote share (of **A**) is defined as follows:

$$f_H = \frac{1}{n} \sum_{i=1}^n E[X_i^n \mid H]. \quad (7)$$

$$f_L = \frac{1}{n} \sum_{i=1}^n E[X_i^n \mid L]. \quad (8)$$

Specifically, f_H (f_L , respectively) is the excess expected vote share of **A** conditioned on world state H (world state L , respectively).

THEOREM 3. Given an arbitrary sequence of instances and arbitrary sequence of strategy profiles $\{\Sigma_n\}_{n=1}^\infty$, let f_H^n and f_L^n be the expected vote share for each Σ_n . Let

$$F = \liminf_{n \rightarrow \infty} \sqrt{n} \cdot \min \left(f_H^n - \frac{1}{2}, \frac{1}{2} - f_L^n \right)$$

- If $F = +\infty$, the fidelity of Σ_n converges to 1, i.e., $\lim_{n \rightarrow \infty} A(\Sigma_n) = 1$.
- If $F < 0$ (including $-\infty$), $A(\Sigma_n)$ does NOT converge to 1.
- If $F \geq 0$ (not including $+\infty$), and the variance of $\sum_{i=1}^n X_i^n$ is $\Theta(n)$ under every world state, $A(\Sigma_n)$ does NOT converge to 1.

REMARK 1. The original theorem in [29] is presented on excess expected vote share, which is the expected vote share of the informed majority decision minus the majority rule's winning threshold. For the convenience of our presentation, we translate the theorem to expected vote share of A while keeping it equivalent to the original theorem.

D Proof of Theorem 1

PROOF. When $\xi \leq \frac{1}{2}$, for each n , let each Σ_n be all the agents unanimously vote for A . According to 1, no agents play a weakly dominated strategy. When the world state is L , the winner is still A , so $\lambda_L^R(\Sigma_n) = 0$, which implies that $A(\Sigma_n) \leq P_H < 1$. Finally, since $\xi \leq \frac{1}{2}$, any deviation group contains at most $\frac{n}{2}$ agents. Consequently, any possible deviating strategy still contains at least $\frac{n}{2}$ votes for A , which guarantees that A is the winner. Since the outcome of the vote cannot be changed, no agent can get an expected utility gain exceeding any $\varepsilon > 0$. Therefore, Σ_n is a $\frac{n}{2}$ -strong equilibrium for any n .

When $\xi > \frac{1}{2}$, we show that a group of majority agents can always successfully deviate to a deviating strategy profile that asymptotically reaches the informed majority decision with probability converging to 1.

Suppose $\{\Sigma_n\}$ is a strategy profile sequence such that $A(\Sigma_n)$ does not converges to 1. This implies that there exists a constant $\delta > 0$ and a infinite set $N_1 \in N_{\geq 0}$ such that for all $n \in N_1$, $1 - A(\Sigma_n) \geq \delta$. Now we construct the deviating strategy sequence $\{\Sigma'_n\}$. We fix an arbitrary n . Let D be a set of majority agents with $D \geq \xi \cdot n - 1$. Given that $\xi > \frac{1}{2}$, $|D| > \frac{1}{2}$ for all sufficiently large n . For all agents not in D , let $\beta_h = \frac{1}{n-|D|} \sum_{i \notin D} \beta_h^i$ and $\beta_l = \frac{1}{n-|D|} \sum_{i \notin D} \beta_l^i$ be the average strategy of all such agents. Now we define the strategy of agents in D . All agents play the same strategy with $\beta_h = \frac{n}{2|D|} + \beta_h \cdot (1 - \frac{n}{|D|}) + \varphi$, and $\beta_l = \frac{n}{2|D|} + \beta_h \cdot (1 - \frac{n}{|D|}) - \varphi$. $\varphi > 0$ is a small constant that still keeps β_h and β_l between $[0, 1]$. This construction averages the two groups of agents described in the sketch and creates a symmetric deviating strategy. With the condition that $|D| > \frac{n}{2}$, it is not hard to show that such φ exists. In this way, we figure out the expected vote share of Σ_n .

$$\begin{aligned} f_H &= \frac{1}{2} + \frac{|D|}{n} \cdot \varphi \cdot \left(\frac{P_{hH}}{P_{hH} + P_{hL}} - \frac{P_{lH}}{P_{lL} + P_{lH}} \right) > \frac{1}{2}. \\ f_L &= \frac{1}{2} + \frac{|D|}{n} \cdot \varphi \cdot \left(\frac{P_{hL}}{P_{hH} + P_{hL}} - \frac{P_{lL}}{P_{lL} + P_{lH}} \right) < \frac{1}{2}. \end{aligned}$$

Both inequality holds because $\Delta > 0$. Therefore, by applying the first case of Theorem 3, the fidelity of Σ'_n converges to 1. Therefore, the majority agents in D get constant utility increase under infinitely many n , and $\{\Sigma_n\}$ fails to be an equilibrium sequence with ε converging to 0. $\square$

E Equilibria in Each Segment

In this section, we display the "optimal" equilibria achieving the maximized ξ^* in each segment as an extension of the proof sketch. In each segment, the equilibrium exhibits a different structure and leads to different features of the segment. Throughout this section, we assume that $P_{lL} \geq P_{hH}$ for a clearer illustration. In each equilibrium described, all majority agents play the same strategy, and all the minority agents in the same group play the same strategy.

We start from $\alpha > \theta$ where an n -strong equilibrium exists. In the equilibrium, majority agents play a modified informative voting such that $\beta_h = 1$ and $\beta_l \geq 0$. Deng et al. [15] show that this majority strategy guarantees an n -strong equilibrium regardless of minority behavior. When $P_{hH} = P_{lL}$, $\beta_l = 0$, and the strategy becomes informative voting that maximizes the margins. When $P_{lL} > P_{hH}$, the H side margin $f_H - \frac{1}{2}$ is more fragile than the L side margin $\frac{1}{2} - f_L$ under informative voting. Therefore, β_l is raised to strengthen the L side in sacrifice of the H side.

In the "Flat" Segment 1, the majority's equilibrium strategy keeps the same pattern of $\beta_h = 1$ and $\beta_l \geq 0$. Moreover, β_l decreases as α decreases, reaching 0 at the transition to Segment 2. The minority agents always vote for A in the equilibrium strategy. Specifically, when $P_{hH} = P_{lL}$, Segment 1 reduces to a point. The flat threshold occurs because the minority A votes offset the margin imbalance caused by signal asymmetry ($P_{lL} > P_{hH}$). As these A votes strengthen the H side margin, the majority can adopt strategies closer to informative voting despite a shrinking population. Strategies closer to informative voting have larger $\beta_h - \beta_l$, which yields a larger sum of two margins $f_H - f_L$. Consequently, the margins remain constant with the majority population and keep the segment flat. At the transition to Segment 2, minority A votes exactly balance signal asymmetry, allowing majority agents to adopt pure informative voting.

In the "Steep" Segment 2, the majority of agents play a modified informative voting, but in another direction: $\beta_h < 1$ and $\beta_l = 0$. Moreover, β_h decreases as α decreases. The minority agents still always vote for A . In this segment, minority A votes overcompensate for signal asymmetry, and the L side margin becomes weaker under informative voting. Therefore, the majority agents must sacrifice total margin to balance both sides, causing the threshold decline.

The "Non-Linear" Segment 3 exhibits the most complex equilibrium structure. The majority of agents follow $\beta_h < 1$ and $\beta_l = 0$, and β_h increases toward 1 as α decreases. Minority agents split into two groups. In the larger group (Group 1), all the agents always vote for A . In the smaller group (Group 0), agents play a strategy of $\beta_h = 0$ and $\beta_l \in (0, 1)$. As α decreases, the strategy of Group 0 gets closer to always voting for R , reaching it at the transition to Segment 4. In this equilibrium, opposing votes from the two groups offset each other's deviation power, maximizing tolerance to deviations. The non-linear relationship arises because both Group 0's strategy and fraction change simultaneously as α varies, creating multiplicative effects on margin changes.

Finally, in the “Tail” Segment 4, all the majority agents vote informatively. Minority agents maintain their two-group structure: the larger Group 1 always votes for **A**, while the smaller Group 0 always votes for **R**. Opposing votes between groups offset their collective deviation power, while the size difference helps balance signal distribution asymmetry. Unlike Segment 3, strategies remain fixed as α changes, restoring the linear threshold relationship.